

TONGHE, China

WMO: 509630

Lat: 45.9825N

Lon: 128.7270E

Elev: 361

StdP: 14.51

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.2	-19.4	-28.3	1.1	-22.8	-24.7	1.4	-19.0	27.5	4.9	25.1	1.0	4.6	250	0.574

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.9	86.6	71.6	84.2	71.3	81.8	70.4	76.2	82.3	74.5	80.7	72.9	78.8	8.2	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
74.3	129.8	79.9	72.6	122.5	78.4	70.9	115.6	76.7	40.1	82.6	38.5	80.8	37.0	79.0	83.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 25.3	22.3	19.3	DB	-29.0	91.2	5.1	2.5	-32.7	93.0	-35.7	94.5	-38.5	95.9	-42.2	97.7	(4)
(5)			WB	-29.2	78.9	5.0	2.3	-32.9	80.6	-35.8	81.9	-38.6	83.2	-42.2	84.9	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	38.1	-4.0	5.5	23.5	42.9	56.6	68.0	72.8	69.4	57.8	41.7	19.9	1.1	(6)
(7)		DBStd	27.94	7.72	9.47	11.15	7.80	7.00	5.80	4.07	4.77	6.45	7.94	11.57	8.62	(7)
(8)		HDD50	6708	1674	1247	820	240	18	0	0	12	277	903	1517		(8)
(9)		HDD65	10364	2139	1667	1285	663	273	35	1	16	228	722	1353	1982	(9)
(10)		CDD50	2367	0	0	0	27	224	541	706	603	246	20	0	0	(10)
(11)		CDD65	546	0	0	0	0	13	126	242	153	12	0	0	0	(11)
(12)		CDH74	3820	0	0	0	13	235	982	1597	881	110	2	0	0	(12)
(13)		CDH80	913	0	0	0	1	48	284	416	155	9	0	0	0	(13)
(14)	Wind	WSAvg	7.9	8.3	8.7	9.3	9.5	8.9	7.5	6.4	6.0	5.8	7.7	8.5	8.4	(14)
(15)	Precipitation	PrecAvg	23.5	0.2	0.2	0.4	1.0	2.2	3.5	6.1	5.4	2.5	1.3	0.4	0.3	(15)
(16)		PrecMax	34.2	0.5	0.6	1.3	2.5	5.3	7.4	15.1	10.4	8.0	3.6	1.5	0.5	(16)
(17)		PrecMin	14.5	0.0	0.0	0.0	0.0	0.4	0.4	2.3	0.3	0.4	0.3	0.0	0.0	(17)
(18)		PrecStd	4.4	0.1	0.1	0.3	0.7	1.3	1.5	2.7	2.7	1.5	0.9	0.3	0.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.5	37.8	55.9	75.3	85.1	90.2	90.3	86.5	81.2	71.4	54.3	34.5	(19)
MCWB			18.9	31.1	44.2	54.4	62.2	69.1	74.1	74.7	68.3	56.9	45.7	31.6	(20)	
2%		DB	16.1	32.1	49.2	69.1	80.3	86.5	87.0	84.0	77.2	64.9	45.8	24.2	(21)	
		MCWB	14.3	27.6	39.7	51.5	60.8	68.1	74.0	73.8	65.8	53.1	39.2	22.1	(22)	
5%		DB	12.5	26.6	44.1	63.8	75.8	83.3	84.6	81.5	73.8	59.6	40.1	19.5	(23)	
		MCWB	11.0	23.6	36.1	49.0	58.8	67.6	73.6	72.1	63.3	49.9	35.4	17.6	(24)	
10%		DB	9.2	22.1	39.8	58.4	71.6	79.8	81.9	79.0	70.7	55.4	35.8	14.8	(25)	
		MCWB	8.0	19.3	33.3	46.0	56.7	66.9	72.8	71.0	61.3	47.8	32.1	13.1	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.4	31.7	46.0	56.2	66.2	74.4	79.4	77.9	71.2	59.1	46.1	30.9	(27)
MCD B			21.3	36.9	53.6	69.3	78.8	84.3	85.2	83.8	77.6	69.3	53.7	35.2	(28)	
2%		WB	14.3	27.6	40.4	52.7	62.9	72.0	76.9	75.6	68.1	54.7	39.8	22.2	(29)	
		MCD B	16.0	31.8	47.9	67.1	74.8	81.1	83.0	81.4	74.4	61.5	44.7	24.1	(30)	
5%		WB	11.0	23.4	36.8	49.8	60.7	70.3	75.4	73.9	65.4	51.4	36.0	17.6	(31)	
		MCD B	12.4	26.7	43.3	61.9	71.5	79.2	81.7	79.2	71.6	57.9	39.8	19.5	(32)	
10%		WB	8.0	19.3	33.7	47.0	58.4	68.6	73.9	72.3	62.9	48.3	32.1	13.3	(33)	
		MCD B	9.3	22.0	39.0	57.0	69.2	76.7	79.9	77.3	68.6	54.9	35.4	14.8	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	20.4	23.7	20.8	21.4	21.1	17.4	14.9	15.9	22.1	20.3	17.7	18.3	(35)
MCDBR			22.9	26.7	24.2	31.4	29.6	23.4	19.8	19.7	26.3	28.0	22.3	21.7	(36)	
MCWBR			21.3	22.4	15.5	16.6	13.4	9.2	9.1	10.3	14.4	16.5	16.3	19.6	(37)	
5% WB		MCDBR	22.7	26.3	22.1	28.4	24.0	18.7	16.0	16.3	21.6	24.9	21.4	21.9	(38)	
		MCWBR	21.1	22.2	14.5	15.8	11.7	8.5	9.1	9.6	13.0	16.0	16.2	19.8	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.327	0.354	0.432	0.534	0.576	0.589	0.561	0.500	0.451	0.458	0.383	0.338	(40)	
taud		2.056	1.985	1.876	1.756	1.729	1.794	1.893	2.042	2.080	1.933	2.027	2.033	(41)		
Ebn at Noon		231	249	245	231	226	225	229	239	239	209	205	208	(42)		
Edh at Noon		32	42	54	67	71	67	60	50	44	44	33	29	(43)		
(44)	All-Sky Solar Radiation	RadAvg	656	1016	1351	1486	1657	1751	1601	1487	1306	906	601	515	(44)	
(45)		RadStd	51	60	103	132	161	207	154	119	103	62	49	28	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+83	N/A
(47) Regional (6 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page